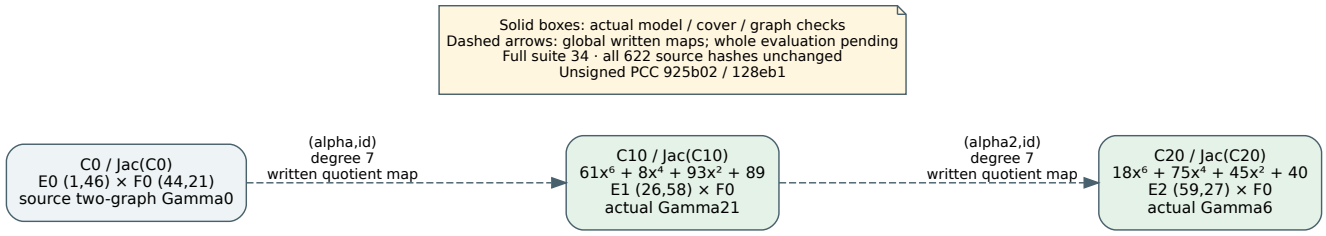
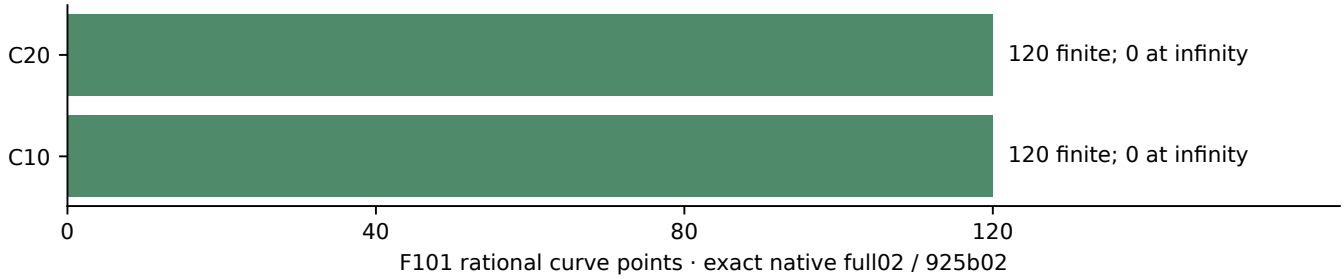
Two unbalanced-transfer models over F101

2026-11-12 · model phase A · native34 / algebra23 / two bound Lean identities

The complete changed suite passes34 tests,zero failures/ignores/filters; oldguards,formatting,strictClippy and all622 source/fixture/engine hashes pass. Full02/925b02 retains whole82,084-byte output as standardXZ9,716, with raw/archive hashes,source snapshots,status and exact roundtrip. Selected01/feff57 is one diagnostic with33 filtered. [pcc:unverified]



Actual equations and transported two-graphs. Both leading and constant coefficients are nonsquares;no normalization or retwist is applied. Solid boxes are native model/cover/graph checks. Dashed arrows are global written quotient maps;whole Jacobian evaluation is phaseB. [pcc:unverified]



Recorded exhaustive F101 curve counts,not sampled values or timings. Both models have N1=120,N2=10436 and Jacobian order12317. Both infinities and both x=0 points exist over the declared extension and are checked even though none is F101-rational. All six Weierstrass roots are checked. [pcc:unverified]

Exact covers and graphs

For f=c6*x6+c4*x4+c2*x²+c0,the direct quotient uses (x²,y) and cubic coefficients[c0,c2,c4,c6];the reciprocal uses(1/x²,y/x³) and[c6,c4,c2,c0]. Short coordinates are X=a*u+b/3,Y=a*v,followed by X→s²X,Y→s³Y.

Cover	Target	a	b/3	s
C10 direct	F0	61	70	43
C10 reciprocal	E1	89	31	43
C20 direct	F0	18	25	33
C20 reciprocal	E2	40	15	33

All coefficient identities and every rational curve point image pass. The source graph is independently extracted from[2]Q0 using actual T0 Cantor pulls and compared with C0′s old root pairs. Actual transported graphs give Gamma21/Gamma6;the four-graph Q20 is A6 up to relative sign. One-factor7 four-transport is Galois stable but fails anti-isometry in 96/256 product-pair comparisons,all nonidentity values-1. This is the credited CT44 obstruction in actual coordinates. [pcc:unverified]

Global proof, formal scope and next maps

Canonical product pullbacks P_i have degree4 and norm[2]. Their exact transported kernels give quotient maps F10 P0=P10(alpha,id) and F20 P0=P20(alpha2 alpha,id),of separable degrees7/49. The adjoint norms descend diag(7,1)/diag(49,1),so they are nonuniform on the full Jacobian. Degree coprimality preserves both109/113 DLP subgroups. The written proof 128eb1 uses actual covers,Picard pullbacks,optimality and the quotient property;it does not cancel maps in finite point groups. [pcc:unverified]

These C10/C20 models are geometrically distinct from the earlier six by their intrinsic factors and canonical sum degree4 versus16. They are two additional explicit presentations inside the fixed72-Jacobian catalog; its previous figures and counts retain their scope. [pcc:unverified]

Faithful local formal checks

UnbalancedCovers.direct_short_quotient/1bde23 and reciprocal_short_quotient/d03d51 prove the universal cleared cubic-to-short identities over every commutative ring. Both statements are exactly bound; gate01 passes two/zero trusted native. The project has171 written declarations in34modules;the new bulk compiled count is still pendingCI. Actual variety/Picard/quotient-degree bridges are written,not LeanAVtheory. [pcc:unverified]

Algebraf404e3 passes23 tests and11 input guards. Seven certificates are emitted;the five old certificates replay byte-identically. New IDC1hdb5d8c25ad6d1262 and IDC1h15a63844aad1bcd0 each have32issuer+64extra checker points over2^61-1 and the final Lean-source digest. Evaluated certificates and local Lean acceptance supply no scientific signature. [pcc:unverified]

Exact remaining implementation

PhaseB must adapt the existing sextic chart to nonsquare leading coefficients over Fp12,compute cover norms from the actual input Mumford divisor,and execute both maps/duals/recovery on all12317 rational classes. The actual7/49 kernels,Frobenius and all exceptional fibres stay required. UNBALANCED_PHASE_B_NOTES.md preserves formulas and falsification controls; the predicted uniform-norm mutant counts have not been executed. [pcc:unverified]

PhaseC must construct T20 from the actual image of Q20 in C20,identify its unique Richelot matching,and check the full primitive-four-glue transfer. Global model existence and graph matching do not complete these maps. No computational advantage,solver cost or timing claim is supplied by this model-construction component. [pcc:unverified]

Sources and reproduction

Sources:Bruin–Doerksen0902.3480 Lemmas2.4/2.6,Definition2.7,Theorem3.2; Galbraith25/26;Milne degree/polarization/quotient results as cited in ISOGENY_CYCLE_PROOF_V2.md;exact externalCT44cc5d5541,original blob retained. Characteristic101 conditions checked;characteristic0 Proposition2.8 unused. KB MCP absent/wrapper unavailable/atlas encrypted;no novelty inference. [pcc:unverified]

Native command:cargo test –release –offline –locked – –nocapture, 15-minute bound plus oldguards/fmt/strictClippy and622 before/after hashes. Rust1.99.1,Lean4.35.0. The focused workflow’s algebra-certificates job replays23 tests and compares seven exact outputs. REPORT.md links raw custody,proofs,statement bindings and full sources. Editable DOT/SVG,recorded counts/renderer and this PDF are retained beside it. Root51d ownCI allfourjobs successful2c4744;consumerf656 fullCI pending. All reported scientific results remain unsigned. [pcc:unverified]